① $V(t) = 200 + t$

$\frac{dQ(t)}{dt} = R_{in} C_{in} - R_{out} C_{out}(t)$ $\qquad C_{out}(t) = \frac{Q(t)}{V(t)}$

$Q' + \frac{R_{out}}{V(t)} Q = R_{in} C_{in}$ $\qquad \frac{R_{out}}{V(t)} = \frac{2}{200+t}$ $\qquad \mu(t) = e^{\int \frac{2}{200+t}\,dt} = (200+t)^2$

$R_{in} C_{in} = 3(1) = 3$

$(200+t)^2 Q = \int 3(200+t)^2\,dt$

$= (200+t)^3 + C$

$Q(t) = 200 + t + \frac{C}{(200+t)^2}$

$Q(0) = 200 + \frac{C}{200^2} = 100$

$C = -4000000$

$Q(t) = 200 + t - \frac{4,000,000}{(200+t)^2}$

$t_{limit} = 300$ $\quad$ b/c $V_{limit} = 500 = 200 + t_{limit}$

$Q(300) = 500 - \frac{4,000,000}{500^2} = \boxed{484 \text{ lb of salt}}$

$C(t) = \frac{Q(t)}{V(t)}$ $\qquad C(300) = \frac{Q(300)}{V(300)} = \frac{484}{500} = \boxed{0.968 = C_{overflow}}$

$C(t) = 1 - \frac{4000000}{(200+t)^3}$ $\qquad \lim_{t\to\infty} C(t) = 1$ $\qquad \boxed{1 > 0.968}$

② $y' = -y^3$

$\frac{dy}{y^3} = -dx$

$-\frac{1}{2} y^{-2} = -x + C_1$

$y^{-2} = 2x + C_2$ $\qquad C_2 = -2C_1$

$y(0) = y_0 = C_2^{-\frac{1}{2}}$

$C_2 = y_0^{-2}$

$y^{-2} = 2x + y_0^{-2}$

$= 2x + \frac{1}{y_0^2}$

$y^{-2} = \frac{2xy_0^2 + 1}{y_0^2}$

$y = \frac{y_0^4}{(2xy_0^2+1)^2}$ $\qquad |y| = \frac{|y_0|}{\sqrt{2xy_0^2+1}}$

$2xy_0^2 + 1 > 0$

$2xy_0^2 > -1$

$x > \frac{-1}{2y_0^2}$

check when $y_0 = 0$:

$y = \frac{0^0}{1} = 0$

$$y(x) = \begin{cases} 0, & y_0 = 0, \ (-\infty, \infty) \\ \frac{y_0}{\sqrt{2xy_0^2+1}}, & y_0 \neq 0, \ (-\frac{1}{2y_0^2}, \infty \end{cases}$$

③ $y' + 2t^{-1}y = y^3 t^{-2}$ $\quad \alpha = -2$ $\quad v = y^{-2}$

$y = v^{-\frac{1}{2}}$

$y' = -\frac{1}{2} v^{-\frac{3}{2}} v'$

$v^{\frac{3}{2}} \cdot \left(-\frac{1}{2} v^{-\frac{3}{2}} v' + 2t^{-1} v^{-\frac{1}{2}}\right) = v^{-\frac{3}{2}} t^{-2} \cdot v^{\frac{3}{2}}$

$-\frac{1}{2} v' + 2vt^{-1} = t^{-2}$

$v' - 4vt^{-1} = -2t^{-2}$ $\quad \mu(t) = e^{-4\int t^{-1} dt} = t^{-4}$

$t^{-4} v = \int -2t^{-6} dt + C$

$t^{-4} v = \frac{2}{5} t^{-5} + C$

$v = \frac{2}{5} t^{-1} + Ct^4$

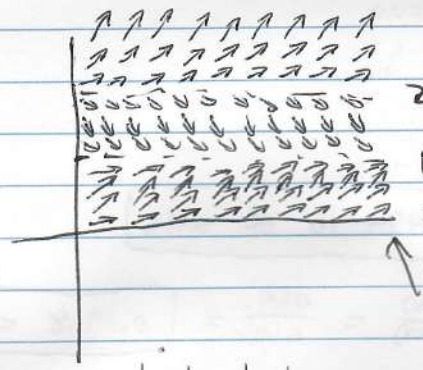
$y^{-2} = \frac{2}{5} t^{-1} + Ct^4$

$\boxed{y = \left(\frac{2}{5} t^{-1} + Ct^4\right)^{-\frac{1}{2}}}$, considering another possibility:

$\boxed{y(t) = 0}$

④ $\frac{dy}{dt} = 0 \quad \forall\, y \in \{0, 1, 2\}$

direction / slope field:



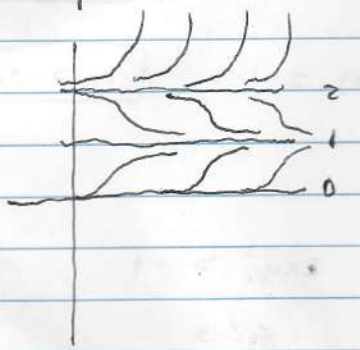
slope decreases as y approaches 0 & 1 increases as the y moves towards & the middle.

The opposite happens between $y = 1$ & 2, where $\frac{dy}{dt}$ is negative, approaches 0 as y approaches 1 & 2, and decreases (or increase in magnitude) as y approaches the middle.

(note the max is not at the exact middles, not 0.5 or 1.5)

$\frac{dy}{dt}$ then increases indefinitely above $y = 2$.

solution curves:

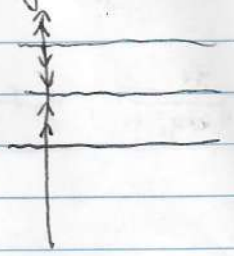


$y = 0$: unstable eq., $\frac{dy}{dt}$ diverges, the negative region isn't drawn b/c $y_0 \geq 0$.

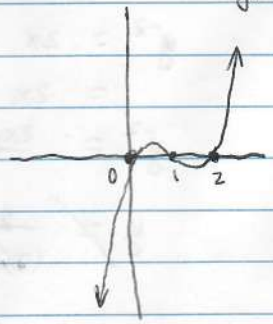
$y = 1$: stable eq. $\frac{dy}{dt}$ converges.

$y = 2$: unstable eq. $\frac{dy}{dt}$ diverges.

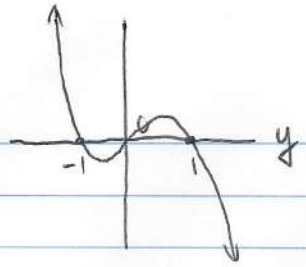
phase line:

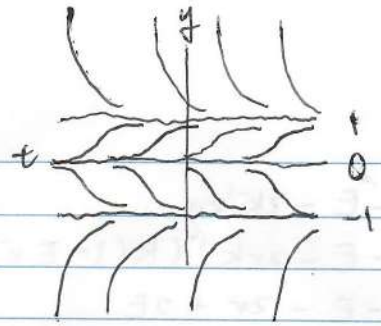


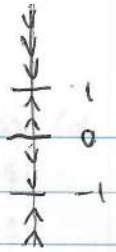
graph of $f(t) =$



⑤ $f(y)$:

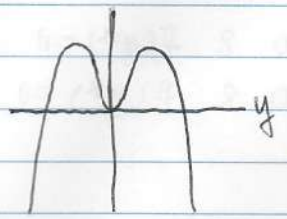


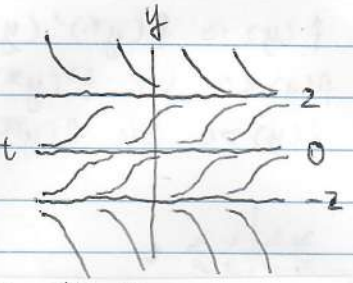


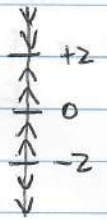


0: unstable, diverges
1: stable, converges
-1: stable, converges

⑥ $f(y)$:







0: semistable 0^- approaches, 0^+ diverges
2: stable, converges
-2: unstable

⑦ a) $0 = ry(1-\frac{y}{k}) - Ey$

$E = r(1-\frac{y}{k})$ $y=0$

$k(1-\frac{E}{r}) = y_2$

given $E<r$, $\rightarrow \frac{E}{r}<1 \rightarrow 1-\frac{E}{r}>0 \rightarrow y_2>0$

b) $y'(-1) = -r(1+\frac{y}{k}) + Ey$ X

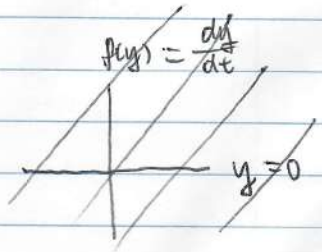
$f(y) = ry - ry^2k^{-1} - Ey$

$f'(y) = r - E - 2rk^{-1}y$

~~$f'(0) = r-E-2rk^{-1}$~~

$f'(0) = r-E$ given $r>E \rightarrow f'(0)>0$

~~$\rightarrow \frac{d^2y}{dt^2}>0$~~

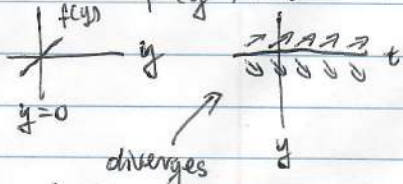


linear approx. $f(y) \approx f(y^*) + f(y^*)'(y-y^*)$ $f(y^*)'>0$

$f(y) \approx f(y^*)'(y-y^*)$ $f(y^*)=0$

$y^*=0 \rightarrow$ if $y>y^*$: $f(y)>0$ b/c $f(y^*)'>0$

if $y<y^*$: $f(y)<0$ b/c $f(y^*)'>0$



$\therefore$ y_1 unstable

⑦ b) cont.

$f'(y) = r - E - 2k^{-1}ry$

$f'(y_2) = r - E - 2rk^{-1}(K(1 - Er^{-1}))$

$= r - E - 2r + 2E$

$= -r + E = E - r$

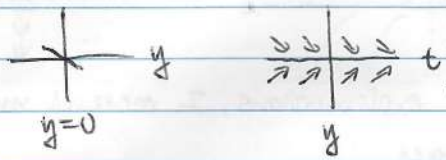
$\therefore E < r$

$\therefore f'(y_2) < 0$

linear aprox. $f(y) \approx f(y^*)'(y - y^*)$

$y^* = y_2 \rightarrow$ if $y > y^*$: $f(y) < 0$ b/c $f^*(y^*)' < 0$ & $f(y^*) = 0$

if $y < y^*$: $f(y) > 0$ b/c $f(y^*)' < 0$ & $f(y^*) = 0$



↖ converges ∴ y_2 stable

c) $Y = Ey_2$

$= E(K(1 - \frac{E}{r}))$

$\boxed{Y(E) = EK - E^2Kr^{-1}}$

d) $Y' = K - 2EKr^{-1} = 0$

$2Er^{-1} = 1$

$\boxed{E = \frac{r}{2}}$